Expressions and equations

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 8.2 × 10

2. 800 ÷ 100

3. Double 8.5

4. Half of 48

5. Solve 1.5x = 4.80.

6. Expand 4(x + 5).

7. Find 0.25 × 0.02.

8. Find 288 ÷ 24.

9. 12 shells are 30% of a collection. Find the whole collection.

10. Red : blue is 3 : 5. How many blue tiles go with 9 red tiles?

11. 4 × 6

12. A box is 4 cm by 6 cm by 3 cm. Find its volume.

13. Two whole numbers total 17. One is 5 more than the other. Find both numbers and show a way to check.

14. A packet contains 7 cards. Write the number in p packets plus 2 loose cards; evaluate at p = 6.

15. Evaluate 2t + 7 when t = 6. Show the multiplication first.

16. Evaluate $2^2 + 7$. Explain what the exponent means.

17. Expand 2(x + 7) and check at x = 6.

18. Simplify 7y + 2y + y. Evaluate at y = 2.

19. A rectangle has height 2 and width z + 7. Write its area two ways.

20. Solve x + 7 = 13. Show a change to both sides.

21. Solve 2x = 12. Check using multiplication.

22. Solve $\frac{2}{3}$ x = 12. Draw 3 equal parts.

23. 7 identical pens cost $14. Write and solve an equation for price p per pen.

CHALLENGE / Explain your plan and check the result

C1. A club buys 8 notebooks at n dollars each and returns 2. It also buys pens for $9. Its final bill is $33. Write two equivalent bill expressions, find n, and check the price against the original purchase and return.

C2. A number machine multiplies by 3, then adds 5. A second machine adds 5, then multiplies by 3. Compare outputs for input 4. How much greater is the second output for any input? Explain with expressions.